

Morphology-Aware ECG Denoising Using Deep Neural Networks for Arrhythmia Detection

Undergraduate Thesis (ongoing for publication)

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Research Problem

- ECG recordings from ambulatory and wearable devices are frequently corrupted by **baseline wander, muscle artifact, and electrode motion artifact**.
- Removing this noise is difficult because parts of the noise overlap with diagnostically important ECG components.
- Aggressive denoising can produce a visually cleaner signal while distorting the **P wave, QRS complex, ST segment, T wave, and R-peak morphology**.
- My thesis therefore focused on a specific question: **Can ECG noise be removed while preserving the morphology required for downstream cardiac analysis?**

What I developed

I designed **MorphoUFormer**, a one-dimensional morphology-aware ECG denoising network. The model combines multiscale convolutional feature extraction with an encoder-decoder structure, gated skip connections, a Transformer bottleneck for longer-range temporal context, and a residual refinement stage. The network first reconstructs a coarse clean ECG and then learns a correction to recover remaining waveform details.

A major part of the work was also the training objective. Instead of optimizing the network only with a conventional point-wise reconstruction error, I developed a **multi-component morphology-preserving loss**. It combines time-domain reconstruction, spectral consistency, first- and second-order temporal derivatives, local structural similarity, and global waveform-shape similarity. The intention was to penalize not only amplitude error, but also changes in QRS slopes, local curvature, frequency content, and overall ECG shape.

Data & Experimental Setup

- **Clean ECG sources:** MIT-BIH Arrhythmia Database and QT Database.
- **Noise source:** MIT-BIH Noise Stress Test Database.
- Used three realistic ECG artifacts: **baseline wander (BW), muscle artifact (MA), and electrode motion (EM)**.

- Generated **eight clean/single/mixed noise conditions** across input SNR levels from **-6 dB to 18 dB**.
- Other dataset used : Icentia11k, Georgia 12 lead ecg dataset, QT dataset.

Evaluation

An important part of the thesis was that I did not evaluate the denoiser only by asking whether its output looked closer to a clean reference.

On the MIT-BIH-based test set of **35,392 ECG segments**, MorphoUFormer increased the mean SNR from approximately **5.96 dB to 14.79 dB**, with a mean SNR improvement of **7.56 dB**.

Against the evaluated deep-learning baselines, it achieved the best result in most reconstruction and morphology-related measures, including PRD, RMSE, MSEWPRD, WDI, output SNR, cosine similarity, and PCC.

For **R-peak detection**, I used 200 records from the large-scale Icentia11k single-lead ECG dataset. After denoising, R-peak detection recall increased from **69.98% to 87.77%**, while F1-score increased from **81.24% to 91.84%**. For a second downstream experiment, I used Lead II recordings from the **Georgia ECG dataset** to compare arrhythmia classification using raw ECG against MorphoUFormer-denoised ECG. With the same classifier architecture, data split, seed, and training protocol, denoising increased classification accuracy from **92.6% to 94.1%** and macro-F1 from **75.4% to 83.6%**. The largest class-level improvement occurred for the minority **AF/AFib** category, where F1 increased from **37.50% to 68.97%**.

The most important result for me was not simply that the denoised waveform had a better SNR. It was that improvements could also be observed in **beat detection and arrhythmia classification**. This reinforced the idea that preprocessing for biomedical AI should be evaluated according to what clinically relevant information survives the preprocessing step, rather than according to signal-level error alone.